Improving Data Centre Reliability for AI Assisted Learning in Cambodia

A synthetic case study for a Cambodian public university digital learning service

Item	Details
Case organization	Tonle Digital Learning Centre fictionalized academic case
Industry	Higher education information technology and data centre services
Project method	Lean Six Sigma DMAIC and operations research decision analysis
Group structure	Four person team with one primary CTQ per member
Dataset	120 synthetic daily records 40 synthetic VOC entries 20 synthetic MSA audits
Study periods	Baseline 1 June to 30 July 2026 Pilot 1 August to 29 September 2026
Submission stage	Stage one project report and data package
Prepared for	Operations Research and Lean Six Sigma course

Data statement Every observation in the project dataset is synthetic. The fictional organization and results are designed for academic DMAIC practice and must not be presented as evidence about a real Cambodian university.

Executive Summary

This project recommends a practical topic for a four-person Cambodian Lean Six Sigma assignment: improving the reliability of a university data centre that supports online classes and an AI learning assistant. The subject connects data-centre operations, artificial intelligence, and Cambodian education while giving each member a separate measurable Critical-to-Quality requirement. The fictional case organization is the Tonle Digital Learning Centre, a small campus service operating a learning platform, identity service, and shared AI application gateway.

The baseline contains 60 synthetic daily observations. Mean availability was 98.327%, mean incident recovery time was 31.4 minutes, mean login success was 96.561%, and the mean 95th-percentile AI response time was 4.25 seconds. Across four CTQ opportunities per day, the baseline produced 162 defects in 240 opportunities, equal to 675,000 DPMO and an estimated 1.05 sigma level with the conventional 1.5 sigma shift. Only 1.7% of days met all four CTQs on the first pass.

The improvement package combines UPS failover, a secondary network path, automated alerting and a standard incident runbook, login health checks, and a priority AI queue with cache warm-up. In the 60-day synthetic pilot, availability rose to 99.881%, recovery time fell to 9.7 minutes, login success rose to 99.638%, and AI response time fell to 2.37 seconds. The pilot produced one CTQ defect in 240 opportunities, or 4,167 DPMO, and 98.3% of days met all four CTQs. Welch tests found $p < 0.001$ for every before-and-after comparison. These results validate the synthetic pilot scenario; they do not prove that a real university would obtain the same effect.

CTQ	Specification	Baseline mean	Pilot mean	Change
Learning service availability	>= 99.5%	98.327%	99.881%	+1.554 points
Incident recovery time	<= 30 min	31.4 min	9.7 min	-69.1%
Login success	>= 98.0%	96.561%	99.638%	+3.077 points
AI response time	<= 4.0 sec	4.25 sec	2.37 sec	-44.2%

Recommended Presentation Claim

The project demonstrates how a small Cambodian university IT team could protect learning time by reducing preventable digital-service failures. The correct claim is that a synthetic DMAIC pilot indicates a credible improvement path. The report does not claim that a real university has already implemented the controls.

Data Integrity and Academic Use

The lecturer permits realistic fictitious data when real data is unavailable. This report uses that permission openly. The fictional organization, all VOC entries, operational observations, audit results, costs, and pilot outcomes are synthetic. The raw CSV includes a data_origin field on every row and uses a fixed random seed of 20260930 so the dataset can be regenerated consistently.

What the Dataset Contains

- 120 daily operating records: 60 baseline days and 60 improved-pilot days.
- 40 synthetic VOC records embedded in the first 40 baseline rows.
- 20 synthetic measurement-system audit checks distributed across both phases.
- Four CTQ defect flags, a total-defect field, and a first-pass all-CTQs-met field.
- Cause variables for power events, network congestion, delayed alerts, cold cache, overload, and peak concurrent users.

Rules for Responsible Submission

1. Keep the synthetic-data statement in the report and presentation.
2. Replace the role labels Member A through Member D with the actual group members before submission.
3. Do not rename the fictional case organization to a real university unless the group collects and verifies real data.
4. If the group gathers real observations later, retain this file as a pilot model and rebuild the analysis from the real data.

Contents

Section	Purpose
Introduction	Cambodian context and topic rationale
Define	VOC, CTQs, charter, scope, SIPOC, process and roles
Measure	Data plan, MSA, baseline, defects, capability and FMEA
Analyze	Pareto, Lean wastes, root causes and statistical validation
Improve	Solution selection, implementation plan and pilot validation
Control	SPC, control plan, reaction rules, ownership and audit
Appendices	Data dictionary, formulas, jamovi steps and question preparation

Introduction

Cambodian Context

Cambodia's Digital Economy and Society Policy Framework 2021 to 2035 identifies digital foundations, digital adoption, and digital transformation as national priorities. The Ministry of Education Youth and Sport also publishes policy material on ICT in education and digital education. Reliable campus infrastructure is therefore a practical operations problem: online learning and AI-assisted study cannot help students when the identity service, learning platform, or application gateway is unavailable or too slow.

The Telecommunication Regulator of Cambodia reported 20,858,249 internet subscriptions and 118.66% penetration for 2025. Subscription penetration is not the same as unique users, but it shows the scale of digital connectivity on which education services increasingly depend. The project stays at campus level and does not attempt to estimate national learning outcomes.

Topic Selection Rationale

Selection criterion	Assessment	Reason
Cambodian social relevance	High	Protects teaching time and student access to digital learning
Data-centre connection	High	Measures availability, incident recovery, authentication and AI service response
Four individual CTQs	High	One distinct CTQ can be owned by each member
DMAIC coverage	High	Supports VOC, MSA, SPC, capability, root-cause tests, FMEA and control
Feasible pilot	High	A small service can test redundancy, alerting, login checks and queue management
Contribution to knowledge	High	Links AI-in-education discussion to measurable service reliability

Case Organization

Tonle Digital Learning Centre is a fictional small data-centre unit within a Cambodian public university in Phnom Penh. It supports a learning management system, student identity and login services, classroom internet access, and an AI learning-assistant gateway. The service window is 12 hours per day. The project boundary starts when a student or lecturer requests the platform and ends when the response is delivered or an incident is restored.

Define Phase

Voice of Customer

The synthetic VOC sample includes 40 student and lecturer responses. Mean importance was 4.83 out of 5, while baseline satisfaction averaged 2.37 out of 5. Respondents most often prioritized uninterrupted platform access, followed by fast AI answers, first-attempt login, and quick technical recovery.

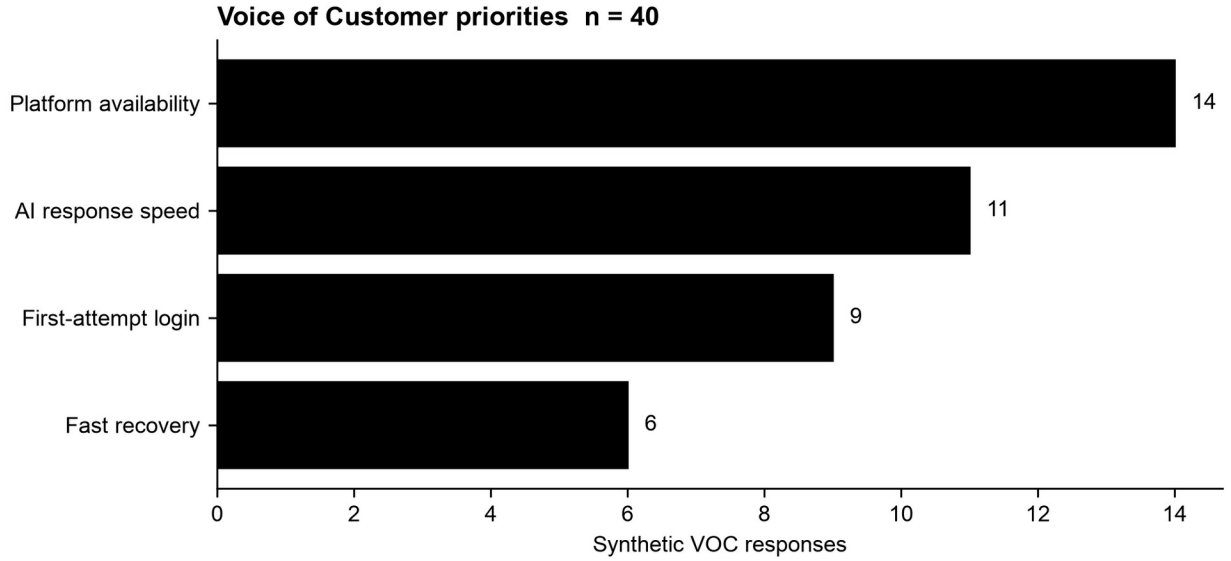


Figure 1 Synthetic Voice of Customer priorities used to derive the four CTQs

CTQ Tree

Customer need	CTQ	Metric	Specification	Owner
The learning platform stays available	Service availability	Available service minutes divided by 720 minutes	>= 99.5% daily	Member A
Service returns quickly after failure	Incident recovery	Minutes from alert to verified restoration	<= 30 min	Member B
Login works on the first attempt	Login reliability	Successful logins divided by attempts	>= 98.0% daily	Member C
The AI assistant answers during class	AI response speed	Daily 95th-percentile response time	<= 4.0 sec	Member D

These four CTQs are independent service requirements. Availability describes whether the service is accessible. Recovery time describes the restoration process after an incident. Login success is a transaction-quality measure. AI response time measures interactive performance. A day can therefore pass one CTQ and fail another.

Project Charter

Charter element	Approved project definition
Business case	Lost learning time and slow AI interactions reduce the value of campus digital services and increase support workload.
Problem statement	In the 60-day synthetic baseline, availability averaged 98.327%, recovery averaged 31.4 minutes, login success averaged 96.561%, and AI P95 response averaged 4.25 seconds. All four means missed their specifications.
Goal statement	Within a 60-day pilot, achieve at least 99.5% availability, recovery within 30 minutes, at least 98% login success, and AI P95 response within 4 seconds.
In scope	Campus data-centre power and network paths, monitoring, incident response, identity/login flow, AI gateway queue and cache, telemetry and daily review.
Out of scope	National education policy evaluation, student grades, AI model accuracy, personal-data profiling, construction of a commercial data centre, and national telecom infrastructure.
Primary customers	Students, lecturers, classroom facilitators and campus IT support staff.
Process owner	Head of Digital Learning Operations fictional role.
Duration	Sixteen weeks for Define through Control, with a 60-day pilot represented in the synthetic dataset.

Team CTQ Responsibilities

Team role	Primary CTQ	DMAIC contribution
Member A project lead	Service availability	Charter, uptime analysis, stakeholder coordination and handover
Member B measurement lead	Incident recovery	Operational definitions, timestamp audit, I chart and runbook
Member C analysis lead	Login reliability	Failure counts, p chart logic, network analysis and health check
Member D improvement lead	AI response speed	Load analysis, queue design, cache warm-up and control limits

SMART Goal Check

- Specific: four named CTQs with one process owner and one team member per CTQ.
- Measurable: every CTQ has a formula, unit, data source and defect rule.
- Achievable: the selected controls address the validated cause variables directly.
- Relevant: the metrics protect learning access and support Cambodia's digital-education direction.
- Time bound: results are evaluated over a defined 60-day synthetic pilot within a 16-week project plan.

SIPOC and Current Process

Suppliers	Inputs	Process	Outputs	Customers
Electricity provider, internet service provider, university IT, identity administrator, AI service team	Power, network links, servers, user accounts, software releases, support staff and monitoring data	Authenticate Route request Process learning or AI request Record telemetry Detect incident Restore service	Available platform, successful login, timely AI response, incident record and service report	Students, lecturers, classroom facilitators and campus management

Detailed Process Map

Step	Activity	Primary risk	Measured output
1	Student opens the learning platform	Service unavailable	Availability
2	Identity service checks credentials	Authentication or network failure	Login success
3	Gateway routes the learning or AI request	Shared queue overload	AI response time
4	Application and AI service return a response	Cold cache or resource saturation	AI response time
5	Monitoring records health and performance	Missing or delayed alert	Detection timestamp
6	Support team diagnoses and restores service	No runbook or unclear ownership	Recovery time
7	Service owner reviews daily CTQ status	Defect not escalated	Control record

Stakeholder Strategy

Stakeholder	Power	Interest	Engagement
University management	High	High	Approve scope and review monthly CTQ dashboard
Digital learning operations	High	High	Own controls and daily reaction plan
Lecturers and students	Low	High	Provide VOC feedback and receive service notices
Internet and power suppliers	High	Medium	Escalate outages against service commitments
Course lecturer	High	High	Evaluate DMAIC application and statistical reasoning

Project Timeline

DMAIC phase	Weeks	Required outputs
Define	1 to 3	VOC, CTQ tree, charter, SIPOC, process map and role assignment
Measure	4 to 7	Data plan, operational definitions, MSA, baseline, stability and capability
Analyze	8 to 10	Pareto, Lean wastes, fishbone, 5 Whys and root-cause tests
Improve	11 to 14	Solution matrix, implementation package, pilot and significance tests
Control	15 to 16	Control plan, SPC limits, reaction rules, training and handover

Project Risks

Risk	Likelihood	Impact	Mitigation
Synthetic results mistaken for real performance	Medium	High	Label each row and every result section as synthetic
Improvement bundle hides which control caused the gain	Medium	Medium	Use a staged real pilot if implementation occurs
Student data privacy issue	Low	High	Collect only service telemetry and aggregate feedback
Control owner does not review alerts	Medium	High	Assign daily ownership and escalation time
AI provider changes response behavior	Medium	Medium	Separate provider latency from campus gateway latency

Define Phase Exit Criteria

- The sponsor accepts the charter and the four CTQ specifications.
- Each group member accepts ownership of one CTQ.
- The process boundary and the difference between customer outcomes and cause variables are clear.
- The team agrees that the dataset is synthetic and suitable only for academic practice until replaced by real observations.

Measure Phase

Data Collection Plan

One row represents one 12-hour operating day. The baseline covers 1 June through 30 July 2026, and the improved pilot covers 1 August through 29 September 2026. Dates are used only to form a realistic sequence. The dataset does not describe actual events. Daily aggregation makes the four CTQs comparable while retaining the event conditions needed for analysis.

Field group	Source in a real project	Sampling frequency	Synthetic implementation
Availability and downtime	Uptime monitor and event log	Continuous with daily roll-up	Generated daily downtime within a 720-minute window
Recovery time	Alert and verified-restoration timestamps	Every incident with daily summary	Generated from power, network, alert and demand conditions
Login reliability	Identity-service transaction log	Every attempt with daily proportion	Binomial failures based on cause conditions
AI response	Gateway request telemetry	Every request with daily P95	Generated from demand, congestion, cache and queue mode
VOC	Anonymous survey	At Define and after pilot	40 synthetic baseline responses
Measurement audit	Manual reference check	20 sampled records	20 paired synthetic audit checks

Operational Definitions

CTQ	Formula and unit	Defect rule	Data type
Availability	100 x (720 - downtime minutes) / 720	Below 99.5%	Continuous proportion
Recovery time	Verified restoration time minus alert time in minutes	Above 30 minutes	Continuous
Login success	100 x successful attempts / total attempts	Below 98.0%	Binomial proportion
AI response	95th percentile of request response time in seconds	Above 4.0 seconds	Continuous

The direction of the specification matters. Availability and login success use lower specification limits because higher values are better. Recovery and AI response use upper specification limits because lower values are better.

Measurement System Analysis

The synthetic audit sampled 20 records. A reference review of downtime produced a mean absolute difference of 0.169 minutes and a maximum difference of 0.56 minutes. Login failure counts agreed in 95.0% of audits. The single mismatch represents a one-transaction classification error and should trigger a log-definition review before real deployment.

Measurement risk	Control	Acceptance rule	Audit result
Clock mismatch	Network time synchronization for gateway, identity service and monitor	Absolute difference <= 1 minute	Mean 0.169 min max 0.56 min
Duplicate login events	Unique transaction ID and retry flag	>= 95% count agreement	95.0% agreement
Inconsistent incident closure	Restoration requires automated health check and owner confirmation	Same closure rule for every incident	Rule defined for the pilot
AI metric inconsistency	Calculate P95 from the same gateway boundary and exclude client rendering time	One query and one time boundary	Rule defined for the pilot

Measurement Decision

The measurement system is adequate for the academic dataset because timestamp differences stay below one minute and login-count agreement meets the 95% minimum. A real project should correct the login classification mismatch and repeat the audit before using the data for performance incentives.

Baseline Performance

Measure	Mean	Standard deviation	Spec	Defect days	One sided Cpk
Availability	98.327%	0.987	>= 99.5%	56	-0.40
Recovery	31.43 min	11.73	<= 30 min	31	-0.04
Login success	96.561%	2.110	>= 98.0%	39	-0.23
AI response	4.254 sec	1.121	<= 4.0 sec	36	-0.08

All baseline Cpk values are below zero because the process means are on the wrong side of their specification limits. This is a capability problem, not merely a stability problem. Tight control around a poor mean would still fail the customer.

Defect Metrics and Pareto

Each day has four CTQ opportunities. The baseline therefore contains 240 opportunities. It produced 162 defects, 675,000 DPMO, 32.5% opportunity yield, and an estimated 1.05 sigma level after the conventional 1.5 sigma shift. First-pass yield, defined as a day meeting all four CTQs, was 1.7%.

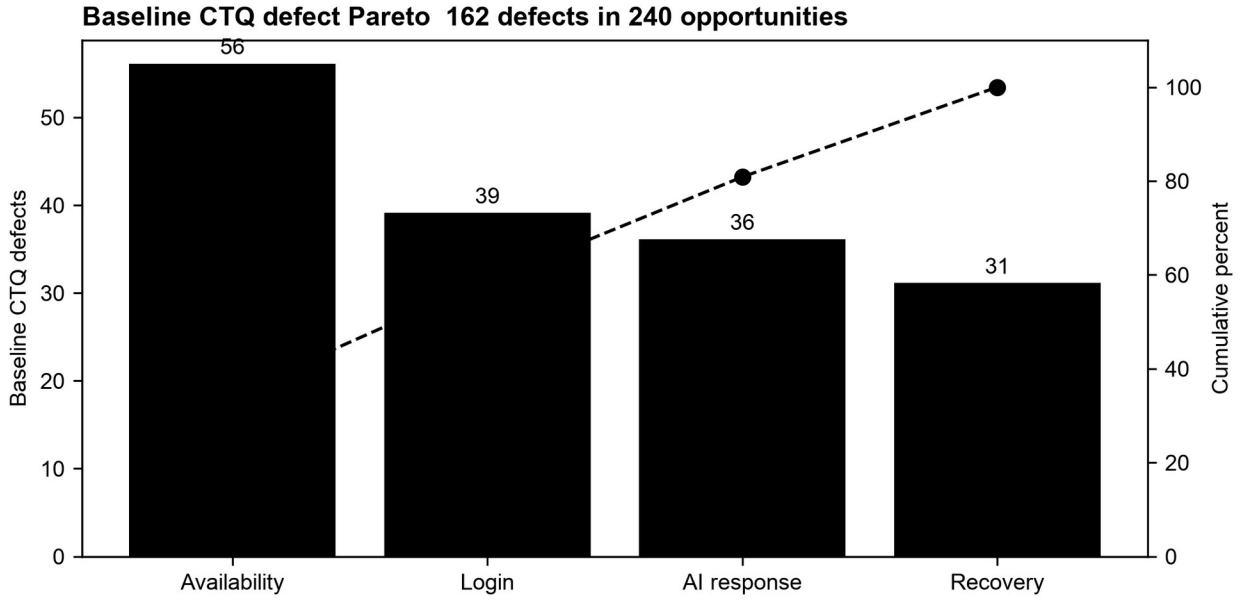


Figure 2 Baseline CTQ defects ranked by frequency with cumulative percentage

Availability generated 56 of 162 defects. Availability and login reliability together generated 95 defects, or 58.6% of the baseline total. The Pareto result supports improvements to infrastructure resilience and network/login handling, but it does not replace root-cause validation.

Baseline Stability and Capability

Individuals and moving-range logic was selected because each daily CTQ produces one aggregated observation. Specification limits and control limits serve different purposes. A specification represents customer acceptance, while a control limit estimates routine process variation. The baseline charts were reviewed before cause testing. High variation coincided with power, network, alerting, and load conditions, so the team treated those variables as candidate special or assignable causes.

Failure Mode and Effects Analysis

Failure mode	Effect	Cause	S	O	D	RPN before	RPN after
Single power path fails	Platform downtime	No UPS failover	10	4	8	320	64
Network congestion	Login and request failure	Single path and no traffic priority	7	6	6	252	72
Alert reaches support late	Long recovery	Manual checking	8	4	7	224	48
AI queue overload	Slow classroom response	Shared queue and cold cache	7	6	5	210	42
Identity service unhealthy	Login failure	No synthetic health check	9	4	5	180	54

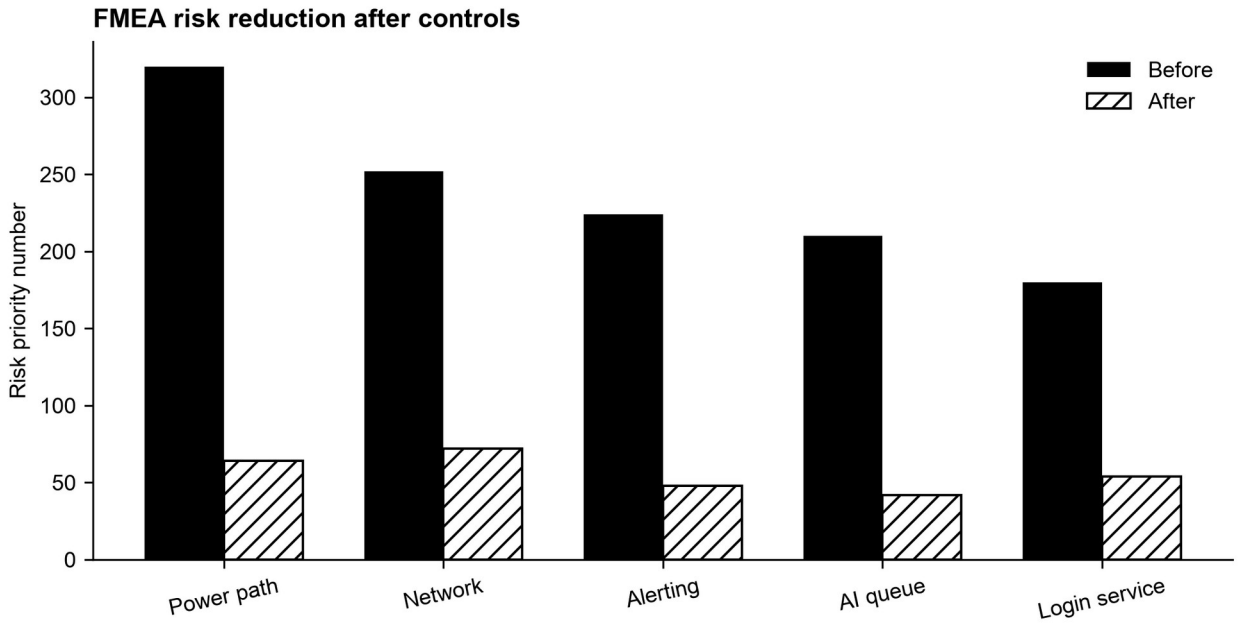


Figure 3 FMEA risk priority numbers before and after proposed controls

Severity was not reduced because the customer consequence of a learning-service failure remains important. The lower post-control RPN values come from reduced occurrence and better detection. RPN is a prioritization aid, not a probability estimate.

Measure Phase Exit Criteria

- Operational definitions are complete and use consistent units.
- The measurement audit meets the academic acceptance criteria.
- The baseline quantifies defects, variation, capability and customer impact.
- The team can trace every reported result to fields in the supplied CSV.

Analyze Phase

Cause and Effect Analysis

Cause family	Potential cause	Expected CTQ effect	Evidence route
Machine	Single power and network path	Lower availability	Compare downtime with and without grid events
Method	Manual or delayed alerting	Longer recovery	Compare recovery by alert mode
Environment	Peak network congestion	Lower login success	Compare login success by congestion state
Machine	Shared AI queue and cold cache	Slower AI response	Relate peak users and AI response
Manpower	No single incident owner	Longer recovery	Review handoff delay and runbook use
Measurement	Inconsistent event closure	Biased recovery metric	Perform paired timestamp audit

Eight Lean Wastes

Waste	Observed in the case	Response
Defects	Failed logins and CTQ failures	Health checks and failure-proof routing
Overproduction	Unneeded repeated retries	Backoff and one clear retry rule
Waiting	Students wait for service restoration or AI response	Failover, queue priority and runbook
Nonutilized talent	Skilled staff perform repetitive health checks	Automated monitoring and clear escalation
Transportation	Not material for digital flow	No action
Inventory	Queued AI requests accumulate	Capacity threshold and queue separation
Motion	Staff switch among separate dashboards	One CTQ dashboard
Extra processing	Duplicate incident entry and repeated authentication	Single incident ID and login health test

Root Cause Validation

Hypothesis	Result	Test conclusion
Grid power events increase downtime	24.47 vs 10.13 min p < 0.001	Validated association in baseline data
Manual or delayed alerts increase recovery	35.46 vs 21.23 min p < 0.001	Validated association in baseline data
Network congestion lowers login success	94.694% vs 97.426% p < 0.001	Validated association in baseline data
Peak users increase AI response time	r = 0.595 R squared = 0.355 p < 0.001	Validated linear association in baseline data

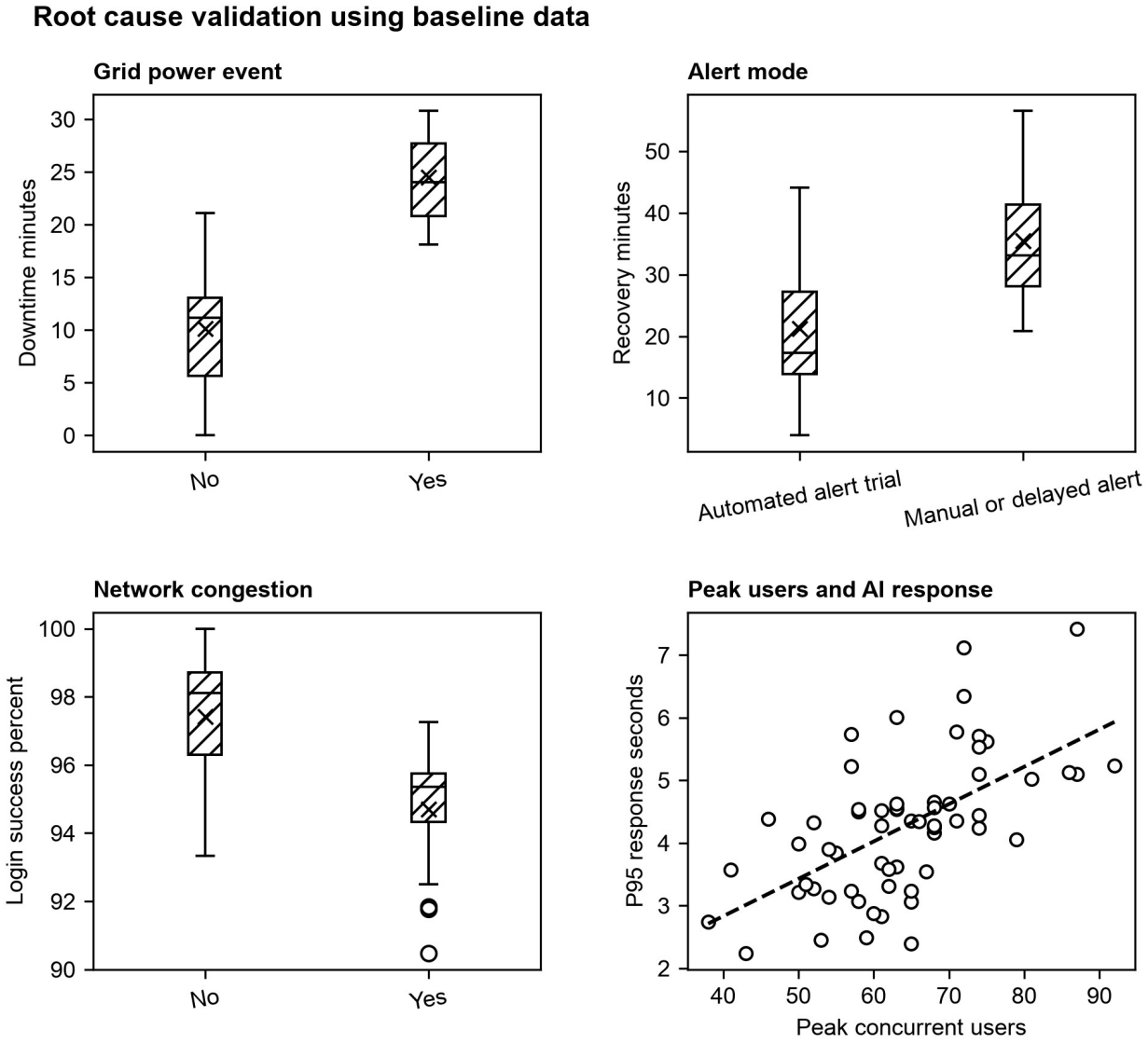


Figure 4 Statistical checks for the four prioritized root causes

The tests support the proposed causal theory but do not prove causation because the baseline is observational and synthetic. A real implementation should stage controls when possible, record the time of each change, and use interrupted time-series or designed experiments to isolate effects.

Five Whys

Level	Availability and recovery chain	Login and AI response chain
Why 1	Why is learning time lost The service becomes unavailable and restoration is slow	Why do students retry or wait Login fails and AI responses exceed the class tolerance
Why 2	Why unavailable Power or network events interrupt a single path	Why does performance degrade Congestion and a shared queue raise transaction delay
Why 3	Why slow to restore Alerts are manual or delayed	Why does load create delay The gateway lacks traffic priority and cache warm-up
Why 4	Why manual Monitoring and response ownership were not standardized	Why no protection The pilot service was designed for functionality before reliability
Why 5	Why not standardized No CTQ-based service control plan existed	Why no CTQ design Customer requirements were not translated into measurable limits
Root cause	Insufficient resilience and incident-control design	Insufficient transaction and workload-control design

Prioritized Root Causes

1. Single infrastructure paths allow external power and network events to become customer-visible downtime.
2. Manual alerting and unclear response ownership extend incident recovery.
3. Congestion and missing login health checks reduce first-attempt authentication success.
4. A shared AI queue and cold cache make response time sensitive to peak concurrent demand.

Analyze Phase Exit Criteria

Each selected improvement must map to one validated cause, protect at least one CTQ, and include a measurable control. Ideas that only improve appearance, add reporting, or move work without removing a cause are not selected.

Improve Phase

Solution Generation

Option	Cause addressed	Expected CTQ effect	Estimated cost
UPS failover plus automatic service restart	Single power path	Availability and recovery	USD 900
Secondary ISP path plus traffic priority	Single path and congestion	Availability and login	USD 480
Automated monitoring and incident runbook	Delayed alert and unclear ownership	Recovery	USD 250
Login synthetic test before class	Undetected identity fault	Login	USD 150
Priority AI queue and cache warm-up	Shared queue and cold cache	AI response	USD 320

Operations Research Decision Matrix

A weighted decision model compares improvement bundles. Scores use a 1 to 5 scale where 5 is most favorable. Weighted score equals the sum of criterion weight multiplied by option score. Cost is scored inversely so a low-cost option receives a higher score.

Option	CTQ impact 35%	Affordability 25%	Speed 20%	Sustainability 20%	Weighted score
Monitoring and runbook only	3	5	5	4	4.10
Infrastructure redundancy only	4	2	3	4	3.30
Application optimization only	3	4	4	4	3.65
Integrated resilience bundle	5	3	4	5	4.30

The integrated resilience bundle receives the highest score because it protects all four CTQs and creates a control system rather than a one-time repair. The estimated one-time implementation cost is USD 2,100. This is a synthetic planning estimate, not a vendor quotation.

Improvement Implementation Package

Workstream	Action	Owner	Verification
Infrastructure	Install UPS failover and secondary network route with automatic health-based switching	Member A and IT operations	Simulated failover test during a nonclass period
Incident response	Configure alerts, one incident commander, standard diagnosis sequence and 30-minute escalation	Member B	Timed recovery drill
Identity service	Run a synthetic login every five minutes and before scheduled classes	Member C	Alert created after controlled test failure
AI gateway	Separate interactive classroom requests, warm the cache and cap background jobs at peak time	Member D	Load test at 80 concurrent users
Management	Review CTQ dashboard daily and hold a weekly defect review	Process owner	Signed weekly review record

Poka Yoke and Jidoka

- Automatic failover prevents a single detected path failure from requiring manual routing.
- A deployment cannot close unless the login health check and AI response smoke test pass.
- The incident form requires an owner, start time, restoration time and cause code before closure.
- The system stops background AI jobs when interactive P95 response exceeds the warning limit.

Pilot Design

The synthetic pilot keeps the same 12-hour service window and comparable demand variables while changing the infrastructure, alerting and AI queue modes. Sixty daily observations provide a balanced comparison with the 60-day baseline. The pilot is evaluated against both statistical significance and practical CTQ specifications.

Pilot Results

Metric	Baseline	Improved pilot	Practical result	Welch test
Availability	98.327%	99.881%	Meets 99.5% target	t = -12.14 p < 0.001
Recovery time	31.43 min	9.71 min	Meets 30-minute target	t = 13.50 p < 0.001
Login success	96.561%	99.638%	Meets 98% target	t = -10.88 p < 0.001
AI response	4.254 sec	2.373 sec	Meets 4-second target	t = 12.37 p < 0.001

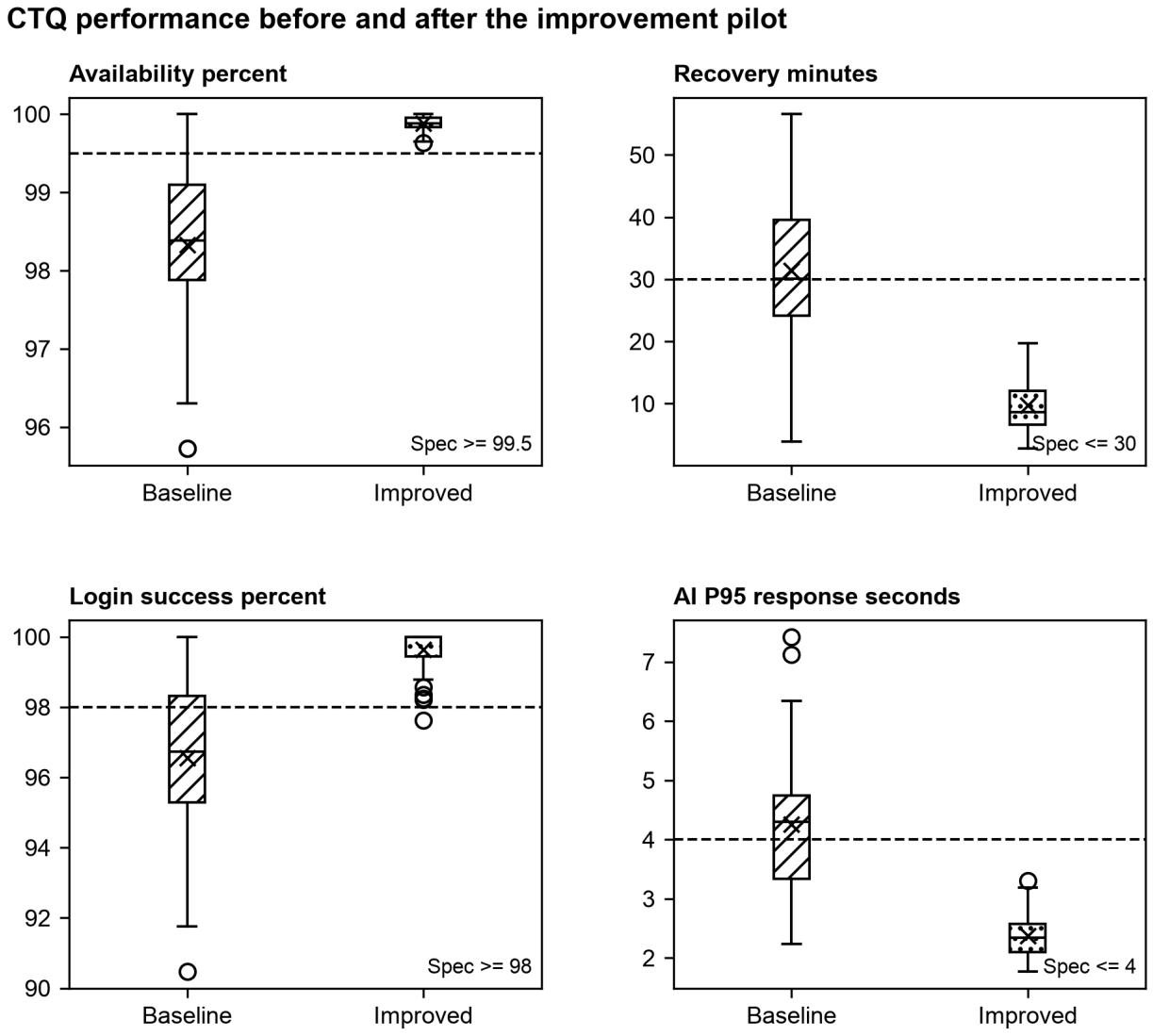


Figure 5 Distribution of each CTQ before and after the synthetic improvement pilot

Capability and Defect Improvement

Measure	Baseline	Improved pilot	Interpretation
Total CTQ defects	162 of 240	1 of 240	99.4% fewer defects
DPMO	675,000	4,167	Lower is better
Opportunity yield	32.5%	99.6%	Four CTQ opportunities per day
Estimated sigma level	1.05	4.14	Includes conventional 1.5 sigma shift
First-pass all CTQs met	1.7%	98.3%	Days with zero CTQ defects

CTQ	Pilot one sided Cpk	Capability assessment
Availability	1.40	Capable against the 99.5% lower limit
Recovery	1.61	Capable against the 30-minute upper limit
Login success	0.93	Meets the mean target but remains below the preferred Cpk of 1.33
AI response	1.49	Capable against the 4-second upper limit

The login CTQ is the only residual weakness. Its mean meets the specification and only one pilot day failed, but the one-sided Cpk of 0.93 shows that day-to-day percentage variation is still too close to the lower specification limit for a mature process. The control plan therefore retains a specific login alert and escalation rule.

Secondary Outcomes

Mean synthetic support tickets fell from 2.92 to 0.73 per day, student satisfaction rose from 2.30 to 4.48 out of 5, and daily energy fell from 52.03 to 49.08 kWh. These are supporting indicators and are not counted as CTQ opportunities.

Improve Phase Exit Criteria

The selected package has a validated link to the prioritized causes, all four mean CTQs meet their specifications, the pilot effect is statistically significant, and the remaining login capability risk has an explicit control.

Control Phase

Control Plan

CTQ	Target	Frequency	Owner	Warning and reaction
Availability	>= 99.5% daily	Continuous daily review	Member A then process owner	If below target, open incident, verify failover and notify management within 15 minutes
Recovery	<= 30 min	Every incident weekly review	Member B then service desk lead	At 20 minutes escalate to infrastructure lead at 30 minutes invoke major-incident process
Login success	>= 98.0% daily	Every transaction preclass check	Member C then identity owner	If preclass check fails, block deployment and switch to backup authentication path
AI response	P95 <= 4.0 sec	Every request hourly roll-up	Member D then application owner	At 3.4 seconds pause background jobs above 4 seconds add capacity or degrade gracefully

Improved Pilot Control Limits

Metric	Center line	Lower control limit	Upper control limit	Specification
Availability	99.881%	99.617%	100.000% operational cap	>= 99.5%
Recovery	9.71 min	0.00 min operational floor	22.75 min	<= 30 min
Login success	99.638%	98.322%	100.000% operational cap	>= 98.0%
AI response	2.373 sec	1.312 sec	3.434 sec	<= 4.0 sec

The plotted limits are descriptive limits from the synthetic pilot, based on the average moving range. Natural limits above 100% are capped at the operational maximum of 100%, and negative time limits are capped at zero. The process owner should recalculate limits after at least 20 stable real observations and should not recalculate them immediately after a defect.

Statistical Process Control

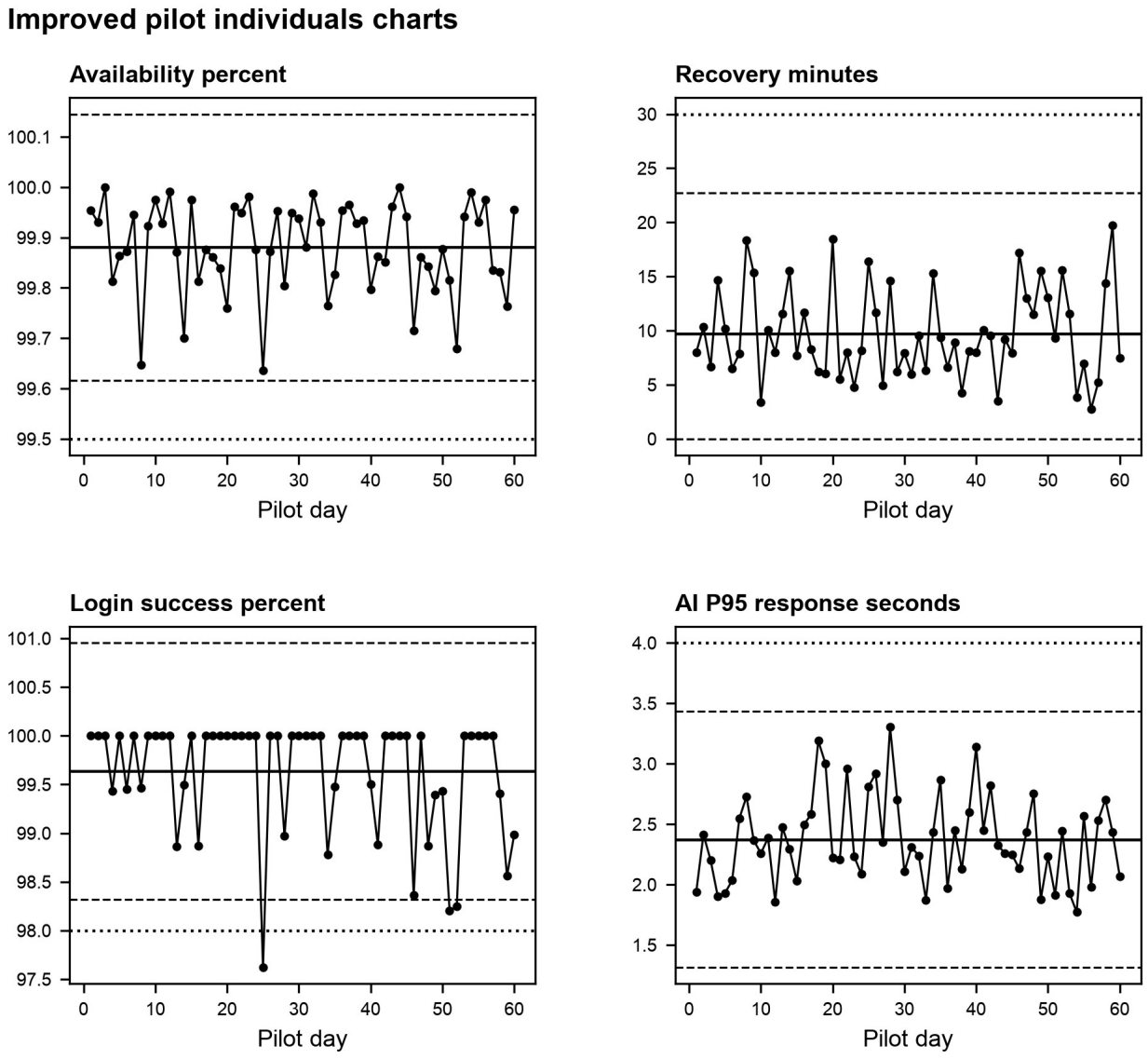


Figure 6 Individuals charts for the four improved-pilot CTQs

Most synthetic pilot observations stay inside their trial control limits. One login-success point falls below both the lower control limit and the 98% specification, matching the single pilot defect. It should be treated as a special-cause signal and investigated, not averaged away. The recovery and AI-response upper control limits are below their customer specifications, and the availability lower control limit is above its specification after operational caps are applied. The login Cpk result and the outlying day make login reliability the continuing control priority.

Reaction Plan

1. Detect the signal and record the affected CTQ, time, user impact and current infrastructure state.
2. Contain the effect by switching paths, pausing background AI work, or using the backup authentication route.
3. Restore the service and verify it with the synthetic login and AI smoke tests.
4. Classify the cause using the agreed cause codes and attach evidence from the relevant logs.
5. Escalate any specification failure, control-limit breach, or repeated warning pattern to the process owner.
6. Review the corrective action at the weekly CTQ meeting and update the runbook only after verification.

Standard Work and Handover

Control activity	Daily	Weekly	Monthly
Review CTQ dashboard	Process owner checks four CTQs	Team reviews trends and defects	Management reviews summary
Test failover	Automated health signal	Review last failover event	Controlled failover drill
Review incident runbook	Use current version	Check incomplete incident fields	Approve revisions
Verify login	Preclass synthetic login	Review failure classifications	Audit account and retry rules
Verify AI service	P95 and queue-depth check	Review peak-hour performance	Reconfirm load threshold

Governance and Ownership

- The process owner accepts the CTQ dashboard, alert rules, runbook and control-limit worksheet.
- Member A transfers availability ownership to infrastructure operations.
- Member B transfers recovery ownership to the service desk lead.
- Member C transfers login ownership to the identity-service administrator.
- Member D transfers AI response ownership to the application-service owner.
- A monthly review confirms whether the control plan remains suitable after workload or provider changes.

Control Phase Exit Criteria

The process owner has accepted the controls, every CTQ has a frequency and reaction rule, control limits are documented, responsibilities are transferred, and the team has defined when the process must re-enter Analyze or Improve.

Benefits and Resource Case

Operational Benefits

Outcome	Synthetic baseline	Synthetic pilot	Difference
Mean downtime per 12-hour day	12.04 min	0.86 min	-11.19 min
Mean support tickets per day	2.92	0.73	-2.18
Mean student satisfaction	2.30 of 5	4.48 of 5	+2.18
Mean daily energy	52.03 kWh	49.08 kWh	-5.7%

Resource Estimate

Item	Synthetic planning estimate
UPS failover and automatic restart	USD 900
Secondary network path and router configuration	USD 480
Monitoring and alert configuration	USD 250
Login health check and validation	USD 150
AI queue and cache configuration	USD 320
Total one-time estimate	USD 2,100

The report does not calculate a financial return because no real tuition revenue, outage cost, staff cost, vendor quotation, or student-loss estimate was provided. A real business case should replace the planning figures with verified local costs and should report a sensitivity range rather than one unsupported return-on-investment value.

Limitations and Ethics

- The complete dataset is synthetic. Statistical significance reflects the data-generation scenario, not real institutional evidence.
- The baseline and pilot are sequential, so a real study would need to check seasonality, course schedules, exams, provider incidents and changes in demand.
- The four interventions are implemented as a bundle. A real staged pilot or designed experiment is needed to estimate each intervention's separate contribution.
- Daily percentages compress transaction-level variation. Real control should also retain event-level logs for diagnosis.
- AI response speed does not measure answer correctness, bias, privacy, academic integrity or learning effectiveness. These require separate governance and research methods.
- No personally identifiable student data is needed for the CTQs. A real project should minimize log retention and restrict access by role.

Interpretation Boundary

The project addresses whether a digital learning service is available, recoverable, accessible and responsive. It does not answer whether AI improves Cambodian students' grades or replaces teaching. Keeping that boundary makes the DMAIC problem operationally testable and prevents an infrastructure study from making educational claims that its data cannot support.

Recommended Real Data Pilot

1. Obtain university approval and assign a real process owner.
2. Collect at least four weeks of baseline event-level telemetry and anonymous VOC feedback.
3. Complete the timestamp and transaction-count measurement audit.
4. Introduce controls in stages where feasible and record every change date.
5. Run the same calculations with real data and compare effect sizes, not only p values.
6. Update control limits after a stable period and preserve all defect evidence.

Conclusion

Improving data-centre reliability for AI-assisted learning is a suitable and distinctive Cambodian DMAIC topic. It connects national digital-education priorities to a small operational process that a four-person group can measure and improve. The project assigns one clear CTQ to each member, uses every DMAIC phase, includes Lean waste analysis and FMEA, validates root-cause associations, applies an operations-research decision matrix, and finishes with a practical control plan.

Within the synthetic case, the integrated resilience package improved all four CTQ means and reduced DPMO from 675,000 to 4,167. First-pass performance rose from 1.7% to 98.3%. Availability, recovery, and AI response became capable against their stated limits; login success met its target but remained the primary capability risk. The correct final recommendation is to use this package as the design for a supervised real pilot, while keeping the synthetic-data disclosure visible.

Project Contribution

The project moves discussion of AI in Cambodian education from a broad opinion question to an operations question with measurable service requirements. It shows that AI access depends on the reliability of ordinary systems such as power, networks, identity services, queues, monitoring and incident response. That connection gives the project social relevance without claiming more than the data can support.

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4. Course reference deck supplied by the lecturer. LSS YB. Applications of Operations Research Lean Six Sigma. 68 slides.

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Source Use Note

The Cambodia sources support the policy and connectivity context only. The course deck informed the DMAIC roadmap. The CloudKH report was treated as a sample structure and was not copied for the organization, dataset, findings or recommendations in this report.

Appendix A Data Dictionary

Field	Definition	Unit or type
record_id	Unique baseline or improved record identifier	Text
date	Synthetic observation date	YYYY-MM-DD
phase	Baseline or Improved Pilot	Category
data_origin	Disclosure that the row is synthetic	Text
service_window_minutes	Scheduled service time per day	Minutes
day_type	Weekday or weekend	Category
exam_period	Synthetic peak academic-demand flag	Yes or No
peak_concurrent_users	Maximum concurrent users during the day	Count
grid_power_event	Whether a grid-power disturbance occurred	Yes or No
network_congestion	Whether high network congestion occurred	Yes or No
alert_mode	Manual delayed alert or automated alert mode	Category
infrastructure_mode	Single-path baseline or resilient pilot design	Category
ai_queue_mode	Shared baseline queue or priority pilot queue	Category
cold_cache_event	Whether AI cache warm-up was absent	Yes or No
overload_event	Whether the phase-specific load threshold was exceeded	Yes or No
downtime_minutes	Unavailable minutes in the 12-hour window	Minutes
availability_pct	Available service minutes divided by 720	Percent
incident_recovery_minutes	Alert-to-restoration elapsed time	Minutes
login_attempts	Total daily login attempts	Count
login_failures	Failed daily login attempts	Count
login_success_pct	Successful attempts divided by attempts	Percent
ai_response_p95_seconds	Daily 95th-percentile AI gateway response	Seconds
energy_kwh	Synthetic daily energy use	kWh
support_tickets	Synthetic daily support-ticket count	Count
student_satisfaction_1_to_5	Synthetic daily satisfaction indicator	1 to 5
availability_defect_lt_99_5_pct	Availability CTQ defect flag	0 or 1
recovery_defect_gt_30_min	Recovery CTQ defect flag	0 or 1
login_defect_lt_98_pct	Login CTQ defect flag	0 or 1
ai_response_defect_gt_4_sec	AI-response CTQ defect flag	0 or 1
total_ctq_defects	Sum of the four daily defect flags	0 to 4
first_pass_all_ctqs_met	One when all CTQs pass	0 or 1
voc_response_id	Synthetic VOC identifier where present	Text
voc_primary_need	Primary need reported in synthetic VOC	Category
voc_importance_1_to_5	Synthetic importance score	1 to 5
voc_satisfaction_1_to_5	Synthetic VOC satisfaction score	1 to 5
msa_audit_selected	Whether the row was selected for the synthetic audit	Yes or No

Field	Definition	Unit or type
msa_reference_downtime_minutes	Synthetic reference downtime in the audit	Minutes
msa_absolute_difference_minutes	Absolute system-reference difference	Minutes
msa_reference_login_failures	Synthetic reference recount	Count
msa_login_count_match	Whether failure count matched the reference	Yes or No

Appendix B Formulas

Measure	Formula
Availability	100 x (service window minutes - downtime minutes) / service window minutes
Login success	100 x (login attempts - login failures) / login attempts
DPMO	total CTQ defects / (records x 4 opportunities) x 1,000,000
Opportunity yield	1 - DPMO / 1,000,000
Estimated sigma level	standard normal inverse of opportunity yield + 1.5
First-pass yield	days with zero CTQ defects / total days
Upper one-sided Cpk	(upper specification - mean) / (3 x sample standard deviation)
Lower one-sided Cpk	(mean - lower specification) / (3 x sample standard deviation)
Individuals-chart sigma	average moving range / 1.128
Individuals-chart limits	center line plus or minus 3 x individuals-chart sigma
FMEA RPN	severity x occurrence x detection
Weighted decision score	sum of criterion weight x option score

Appendix C Jamovi Analysis Steps

1. Open the CSV and set phase, event flags, alert_mode and queue modes as nominal variables.
2. Set availability_pct, incident_recovery_minutes, login_success_pct and ai_response_p95_seconds as continuous variables.
3. Use Exploration then Descriptives to obtain means, standard deviations, boxplots and distributions by phase.
4. Use T Tests then Independent Samples T Test with phase as the grouping variable. Select Welch's test and effect size.
5. For baseline root-cause checks, filter phase to Baseline and compare each metric by its binary cause flag.
6. Use Regression then Correlation Matrix for peak_concurrent_users and ai_response_p95_seconds.
7. Export the tables with the sample size, statistic, degrees of freedom, p value and effect size visible.
8. Interpret statistical results together with the CTQ specifications and the synthetic-data limitation.

Appendix D Question Preparation

Likely lecturer question	Recommended answer
Why did you choose this topic	It connects Cambodia's digital-education direction to an operational problem that four members can measure separately through availability, recovery, login reliability and AI response.
Is the data real	No. Every row is synthetic and labeled as such. The assignment permits realistic fictitious data. The report explains how to replace it with a supervised real pilot.
Why are there four CTQs	The group has four members and the lecturer requires at least one CTQ per person. Each CTQ measures a different customer requirement.
Why use Welch's test	The baseline and pilot are independent groups and do not need equal variances. Welch's test is safer when phase variation differs.
Does p below 0.001 prove the solution works	No. It shows the synthetic groups differ strongly. Causal confidence in a real organization requires controlled implementation and real observations.
Why is login still a concern	Its mean passes and only one pilot day fails, but one-sided Cpk is 0.93, below the preferred 1.33. The process needs continued monitoring and less variation.
What is the difference between control and specification limits	Specifications state customer requirements. Control limits describe expected process variation. A stable process can still be incapable.
How is AI relevant	The CTQ measures whether the AI learning assistant responds quickly enough for class use. The project does not claim that response speed proves learning effectiveness.

Final Submission Checklist

- Replace Member A through Member D with the actual names and confirm one CTQ per member.
- Keep the report, CSV and future slides consistent on all specifications and sample sizes.
- Use only black text and black-and-white figures in the presentation.
- Keep the synthetic-data disclosure on the methods slide and in the presenter notes.
- Practice the eight questions above and explain the remaining login-capability risk honestly.